ELSEVIER

REGULATION OF OVARIAN FOLLICULAR DYNAMICS IN FARM ANIMALS. IMPLICATIONS FOR MANIPULATION OF REPRODUCTION

M. A. Driancourt
Intervet Pharma R&D, BP 67131
49071 Beaucouze Cedex, France

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ABSTRACT

In this review, the main features of folliculogenesis are summarized and compared among species. In the past few years, ultrasonography has clarified follicle growth patterns, and our understanding of follicle maturation has improved considerably. As the follicles develop towards the ovulatory stage, three features appear to be highly conserved across all species: 1) the sequence of events (recruitment, selection and dominance); 2) the sequential need for gonadotropins (FSH for recruitment, LH for dominance) and 3) the large variability of numerical parameters (number of waves per cycle, number of follicles per wave) as well as temporal requirements (time of selection, duration of dominance). In addition, specific follicles may also have variable gonadotropin requirements (thresholds). When patterns of follicle development at different physiological states are compared across species, follicular waves were detected in cattle, sheep and horses and during the prepubertal period in swine, suggesting that ovaries of all species operate on a wave basis unless they are prevented from doing so.

Efficient estrus control treatments should have the ability to affect 1) the wave pattern by preventing the development of persistent dominant follicles containing aging oocytes, and 2) the recruitment of the future ovulatory follicle whatever the stage of the wave at the time of treatment. This would allow synchronous ovulation of a growing dominant follicle. Manipulation of the luteal phase follicular waves after mating or AI may also optimize fertility.

Superovulation is still an efficient technique to obtain progeny from genetically valuable females. Administration of exogenous gonadotropins acts to reveal the underlying ovarian variability. Ovarian response of each female depends on the number of gonado-sensitive follicles present at the time when treatment is initiated. Identification of the number of such follicles for each female would improve efficacy of superovulation, by allocating potential nonresponders to other techniques (OPU/FIV). One of the main components of the within female response to superovulation is the stage of the wave when gonadotropins are injected. Treatment in the absence of a dominant follicle ensures a response close to the female's specific maximum. The development of practical approaches to achieve this still requires further research.

Key words: follicle, maturation, oocyte, gonadotropins, fertility

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The author is grateful to Dr. K. Reynaud for art work.
Correspondance and reprints: Fax 02 41 22 82 51.

E-mail: marc-antoine.driancourt@intervet.com

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INTRODUCTION

Regardless of their physiological stage, the ovaries of all farm species contain a large store of primordial follicles, and more limited populations of preantral (100 to 1000) and antral (50 to 300) follicles. The time required for these follicles to growth from the primordial and preantral stages to ovulation is several months and several weeks, respectively, making them out of the focus of this review. Instead, structural features and regulatory requirements of gonadotropin-sensitive (Gntp- sensitive) and gonadotropin-dependent (Gntp-dependent) follicles will be described and discussed. The Gntp-sensitive follicles become mobilized when exogenous gonadotropins are injected. They measure 0.8 to 2 mm and 1 to 3 mm in sheep and cattle, respectively (43). Gonadotropin-dependent follicles regress when gonadotropin support is withdrawn by hypophysectomy, GnRH antagonist administration, long-term GnRH agonist administration. Follicles become Gntp-dependent when they reach 4, 2, 1, and 10 mm in cattle, sheep, pigs and horses, respectively (43). These two types of follicles were chosen for discussion in this review because 1) ultrasound scanning was developed for all farm species over the past few years such as the mare (95); cattle: 99; sheep: 116; goats: 61), 2) developing associated techniques such follicle aspiration, ovum pick-up, and follicular fluid micropuncture has advanced our knowledge of mechanisms regulating growth of the ovulatory follicles and 3) combining new knowledge of growth patterns and techniques that allow to collect cells at specific stages of folliculogenesis with molecular biology has expanded our understanding of mechanisms regulating follicular maturation. Finally, these follicles are those which are manipulated when estrus control treatments are administered or superovulation attempted.

GROWTH AND TURNOVER OF GONADOTROPIN-DEPENDENT FOLLICLES

Facts and Concepts

Two facts should be emphasized because they affect the mechanisms detailed below. First, for a given female, not all follicles are similar. For example, for a specific size class, there is a 5-fold variation in granulosa cell proliferation in healthy follicles. Furthermore, there is a similar variation between follicles of the same size in their response to a gonadotropin challenge (85). Whether these differences are the result of oogenesis or acquired later during folliculogenesis remains to be clarified. Second, there are large differences among females in follicle populations and gonadotropin concentrations. The former claim has been demonstrated, whatever the species, by the wide (5- to 10-fold) variation among females of the same age and body condition in the number of ovarian follicles. The latter claim is supported by the results of the numerous studies which have failed to relate gonadotropin concentrations to the number of ovulatory follicles (41).

Most early studies characterizing follicular dynamics were limited to the growth phase of the ovulatory follicle. Examination of daily ultrasound scans demonstrated that this phase could be divided in 2 time periods. During the first period, which was closest to luteolysis, the number of follicles growing (forming a cohort) was higher than the number of ovulations (Figure 1). During the second period, which was closest to ovulation, the number of growing follicles was equal to the number of ovulations (Figure 1). This was associated with the presence of a large follicle and the simultaneous regression of the other follicles of the cohort. These observations strongly suggested that the concepts developed by Di Zerega & Hodgen (36) for primates also applied to farm species. More specifically (Figure 1), recruitment was defined as the initiation of gonadotropin-dependent folliculogenesis by a cohort of healthy follicles. Wave emergence is also used to describe this process (65). During the mid-follicular phase, at the time of selection, the number of cohort follicles becomes adjusted to the number of ovulations. As a consequence, a dominant follicle appears, while the other follicles of the cohort regress by atresia. In addition, during the dominance phase, recruitment does not occur (43).

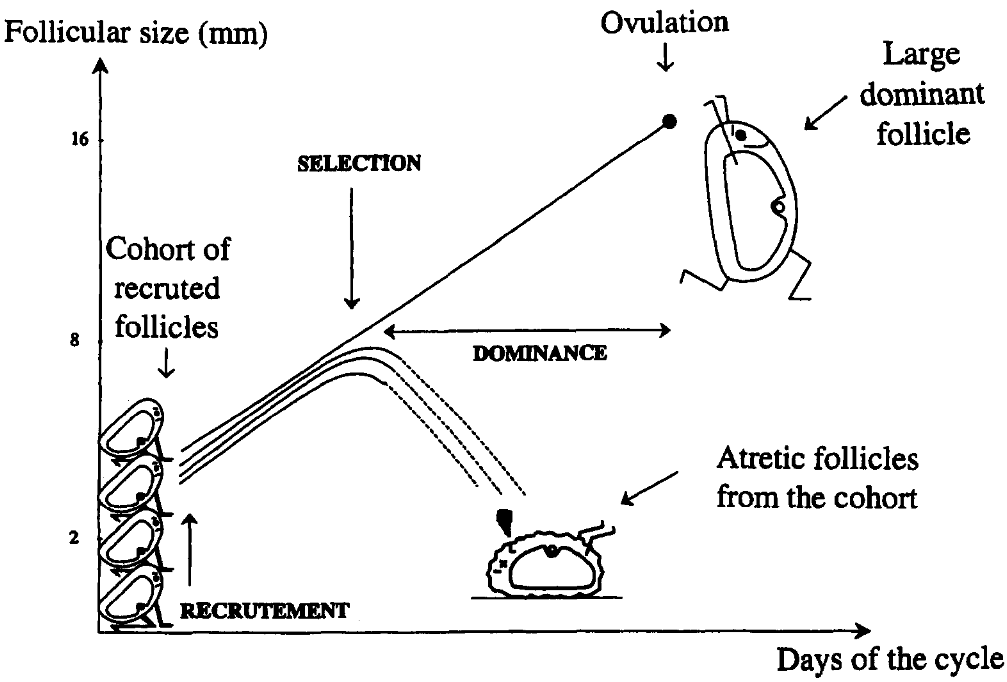

Figure 1. Main events that occur during a follicular wave. At recruitment, a cohort of follicles begins its final growth phase as it enters gonadotropin-dependent folliculogenesis. At selection, the number of growing follicles becomes equal to the number of ovulations. As a consequence, the ovulatory follicle becomes dominant and the other follicles of the cohort regress by atresia (dotted line). On this figure, the ovulatory wave is presented, but the mechanisms are similar in anovulatory waves.

Detailed features of these 3 events (recruitment, selection and dominance) are summarized below. Recruitment of the cohort containing the future preovulatory follicle occurs during a "recruitment window" which lasts 1, 2, or 3 days in sheep, cattle or horses, respectively. To be recruited, follicles need to be healthy; hence, total counts of follicles done to estimate the size of the cohort, are meaningless without getting an estimate of the proportion of those that are able to grow. Only gonadotropin-dependent follicles are recruited. The link between size at recruitment (2mm) and the size at which follicles become gonadotropin-dependent (2mm) is obvious in sheep. This relationship can also be found in primates (2 mm) and mice (0.2 mm). While recruitment and follicle gonadotropin-dependence are associated in most species, the number of recruited follicles growing in the cohort appears to be highly variable between species. It ranges from over 50 in pigs, to 5 to 10 in cattle and 1 to 4 in horses. All follicles of the cohort are potentially capable of ovulating, since selective ablation of all except one will not postpone ovulation in sheep (131) or cattle (57). However, this does not disprove that some follicles may have a competitive advantage in the race towards ovulation.

At selection, the dominant follicle is chosen and the remaining follicles of the cohort become subordinate follicles and enter atresia. This is usually demonstrated by a block in their growth rate followed by a steady decrease in size. The parameters (size, follicle maturation) of the selected follicle have been extensively studied to try to get clues about the cause for survival of a specific follicle. It is generally assumed that the largest follicle of the cohort is likely to be the one selected for ovulation, (65) but the process is very complex, because all the follicles of the cohort are functionally different. The development of a follicular fluid micro-puncture technique in cattle has indicated that estradiol production does not reveal the fate of specific follicles within the cohort (63). In all species, the selected follicle appears to be the first one developing LH receptors on its granulosa cells. Follicles develop LH receptors when they reach 4, 5 to 6, 8 and 25 mm diameter in sheep, pigs, cattle and horses respectively. In all species, there is a close association between these sizes and the sizes at which follicles are selected. In contrast, the intensity of selection (measured by the proportion of the cohort follicles surviving) is highly variable among species. It can be very low in some horses which have a small cohort size, while it is very high for cattle (one selected follicle out of 5 cohort follicles) and swine (12 selected follicles out of a cohort size of 50).

During follicular dominance, preovulatory follicular growth and maturation occur. The other follicles of the cohort complete regression by atresia, while no recruitment occurs. There is a direct relationship between the presence of the dominant follicle and the absence of recruitment, as cauterisation of the dominant follicle immediately induces recruitment (77). The magnitude of dominance is usually defined by the size difference between the dominant follicle and the largest subordinate follicle. Again, very large species differences are detected using this parameter. The size gap may only be 2 to 3 mm in sheep, 3 to 4 mm in pigs, but it will reach about 8 and 15 to 20 mm in cattle and horses, respectively. As a consequence, it is much easier to identify dominant follicles by ultrasonography with high accuracy in the mare than in the ewe.

Follicular Waves throughout Different Physiological Stages

In the previous section, follicular development preceding ovulation was characterized as a sequence of recruitment, selection and dominance. In many farm species, these events are not, however, restricted to the few days preceding ovulation. We will call "follicular waves" the recurrent succession of the sequence of recruitment - selection - dominance throughout time (Figure 2). Features of the follicular waves of specific species and the physiological stages at which they can be observed are detailed below. It is interesting to note that when waves are detected, they appear to develop randomly between ovaries. The only exception to this rule are waves in the early pregnant and the early post-partum cow.

Pre-pubertal period. The only species where follicular development has been carefully characterized by ultrasonic examinations during this period is the bovine (51). Repeated scans of the ovaries of calves or heifers between 2 and 36 weeks of age demonstrated that follicular development occurred in a wave like pattern as early as 2 weeks, with each wave lasting 7 to 9 days. As the animal aged, the diameter of the dominant follicle increased (from 8.5 to 12 mm)(Figure 2). Dominance also became more pronounced, as the size difference between the dominant and largest subordinate follicle increased from 2 to 5 mm. In contrast, growth and regression rates were unaltered.

Although no information is available for young gilts, follicle turnover has been demonstrated in swine during the late pre-pubertal period between Days 140 and 180 of age using laparoscopy repeated every 20 days (70). Changes of the morphological aspects of the ovaries from "honeycomb type" (with large numbers of small follicles) to "grape type" (with several large follicles) were shown to occur within a few days (26). The "grape type" ovaries were not maintained for more than 5 days and these ovaries then reverted to the "honeycomb type"(26). This may be taken as evidence that follicular waves occur in gilts at these ages. However, the "honeycomb type" ovary tended to persist as long as 10 days in some gilts suggesting that waves may be separated by periods when the ovaries remain quiescent.

In sheep, there is evidence suggesting the presence of dominant follicles before puberty in lambs. This was obtained by injecting an ovulatory dose of hCG to see by counting corpora lutea at laparoscopy whether dominant follicles with functional LH receptors were present. Such follicles proved to be present and the number of such follicles was closely related to the number of dominant follicles observed in adult animals differing in prolificacy (39). However, this observation does not demonstrate that regular waves occur at this stage in lambs.

Luteal phase. The number of follicular waves detected throughout the estrous cycle in cattle has been extensively characterized. Interestingly, two (60) or three (114,120) waves per cycle have been described. For 2-wave cycles, recruitment of Wave 1 is identified on the day of ovulation (D0), and by D3, a dominant follicle was present which reached an ovulatory size by D6. It remained static for a few days, until the next wave was initiated around D10 (i.e., D10 after ovulation). The dominant follicle of Wave 2 became the ovulatory follicle. Three-wave cycles were characterized by a shorter static phase of the dominant follicle of Wave 1 and by a longer luteal phase, which allowed time for a smaller Wave 2 dominant follicle to develop (Figure 2). Wave 3 provided the ovulatory follicle. No clear effect of the age or of the breed on the respective proportion of cycles with 2 or 3 waves has been detected. Whether specific animals tend to repeatedly display 2 or 3 waves is not known. In contrast cycles with twin ovulations were never associated with a 2 wave pattern (20). Surprisingly, the relationships between fertility and the wave pattern have not been extensively explored and have generated conflicting results. On one hand, no difference in pregnancy rate was detected between 2 and 3-wave animals (3). On the other hand, low fertility in high producing dairy cows is associated with a high proportion of animals exhibiting 2 wave patterns (JF Roche, personal communication).

Variation in numbers of waves per cycle was also found when follicle waves were detected in other species. Sheep were shown to have 3 or 4 waves, (15, 64), goats 4 or 5 (61) and horses 1 or 2 (59). Swine showed no evidence of large follicles in numbers compatible with a wave during the luteal phase in a histologic study (34) and in response to hCG (MA Driancourt, unpublished). A broad range of compounds (estradiol, inhibin) that show a negative feedback effect on FSH are produced by the porcine CL. Absence of waves may be explained by the lowering of FSH concentrations below the threshold needed for recruitment.

Pregnancy. Repeated ultrasound scanning of the ovaries of pregnant cows revealed that regular emergence of follicular waves was not altered in early pregnant cows (99, 130)(Figure 2).

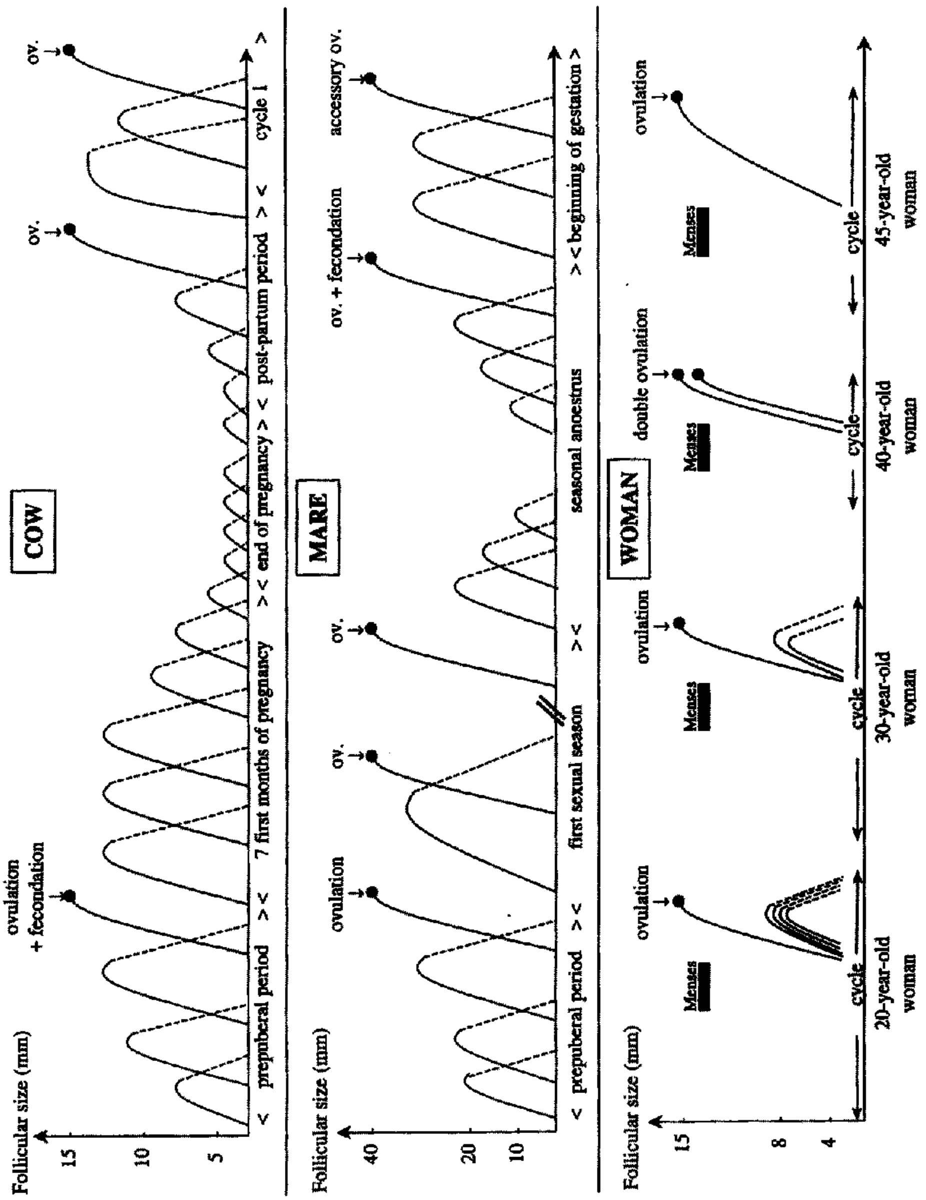

Figure 2. Follicular turnover at different times of reproductive life in 3 models: cow (top panel), mare (central panel), and woman (bottom panel). Growing follicles are presented as continuous lines while atretic follicles appear as dotted lines.

It was noteworthy that the dominant follicle in most waves was preferentially located on the ovary contralateral to the corpus luteum of pregnancy and to the conceptus (99,130). This appears to be due to a local inhibitory action of the conceptus (130). Alterations in the wave pattern were also limited to the later stages of pregnancy (62). The main changes noted were a reduction in diameter (to 9mm) of the dominant follicle after the fourth month of pregnancy. The last month of pregnancy was characterized by a further reduction of the maximum diameter of the dominant follicle (only reaching 6 to 7 mm)(Figure 2). There was a complete cessation of wave patterns during the last 21 days of pregnancy.

Although ultrasonography of ovaries of pregnant ewes is not feasible, there is evidence that the same changes may be occurring in sheep. Large follicles are present up to D50 of pregnancy. Between days 50 and 80, the maximum diameters observed steadily decreased. During the last month of pregnancy, very few follicles larger than 2 mm were observed (40).

Post-partum anestrus. Follicular waves resume early (generally 10 days after calving) during the post-partum period, regardless of the type of cattle studied (dairy or beef) (107,113)(Figure 2). The first ovulation generally occurs later in beef than in dairy cattle. As a consequence, the ovulatory follicle of beef cows originates from a later wave (second, third or fourth) than that of dairy cows (89). This late resumption of ovulation in beef cattle may be caused by the nursing of a calf and/or by an extended period of inadequate energy balance, since cows in poor body condition can have up to 14 waves before first ovulation (126). In sheep, there is also evidence showing that follicular growth quickly resumes during the early postpartum period (15), as most of the ewes given an ovulatory dose of hCG at D10 post partum ovulate (6). The growth pattern of such follicles appears to be already organized into waves similar to those occurring during early pregnancy (15). Interestingly, in both species, evidence supporting a carry-over effect of pregnancy (15,17) suggests a preferential resumption of ovulation in the ovary contralateral to the pregnant uterine horn and CL-bearing ovary.

Seasonal anestrus. Extensive data on follicle turn-over during seasonal anovulatory periods are available in horses and sheep. The main conclusions which emerge are that similar changes appear to occur in both species and that there are large breed and individual differences in the seasonal changes in ovarian function. In some females (saddle-type mares and ewes in shallow anestrus), the wave pattern appears to be maintained (12,125) and large follicles are present throughout anestrus (Figure 2). In sheep, these follicles can be induced to ovulate with an injection of hCG (39). In pony mares and ewes in deep anestrus, follicular waves disappear and follicle diameter does not exceed 10 to 15mm in the pony mare or 3 to 4mm in the ewe (59). The ovaries of these females do not respond to acute administration of LH or hCG. In these species, the wave pattern is slowly re-established during the transition towards the breeding season and the diameter of the dominant follicle slowly increases as the first ovulation approaches (14, 58, 102).

Genetic and Environmental Factors Affecting the Wave Pattern

Wave patterns have been extensively studied in breeds of sheep with large genetic differences in natural ovulation rates due to a single gene (Booroola) or due to several genes (Finn, Romanov). Direct ultrasonographic evidence (13), together with data obtained by the marking of individual follicles with India ink (46) indicate that alterations in follicular growth patterns generating high ovulation rates at the end of the follicular phase can also be observed during the luteal phase (46) or during seasonal anestrus (12). For example, the cohort of follicles growing during Wave 1 of the luteal phase exhibits very similar features as the cohort generating the ovulatory follicles. Number of recruited follicles, their growth rate, the proportion of these which become dominant, and finally the number of dominant follicles are similar at both physiological stages (46). Furthermore, injection of hCG (to ovulate dominant follicles) to anestrous Booroola ewes induces ovulation of a number of follicles typical of the Booroola genotype (39)

Effects of nutrition on follicular waves have also been explored (90, 106). Both studies demonstrated that poor nutrition in cattle was associated with a reduced size of the dominant follicle of all waves, and by a reduced persistence of the dominant follicle of the first wave. In addition, low dietary intake tended to increase the proportion of the cycles with 3 follicular waves. In some countries, cows suffer summer heat stress when temperature and humidity are high. These extreme conditions alter follicular waves to reduce the diameter of the dominant follicle, limit dominance and hasten the turnover of the dominant follicle of the first wave so that emergence of the second wave occurred earlier (8, 137).

MATURATION OF GONADOTROPIN-DEPENDENT FOLLICLES DURING THE FEW DAYS PRECEDING OVULATION

Numerous compounds change concentrations as follicular maturation proceeds (69). In this review, we will only focus on those for which comprehensive information exists on the compound, its receptor, its binding proteins and the proteolysis from binding proteins) exists. Data on rodents will not be included since they are poor models for ovarian function of farm animals. Hence, four main markers of maturation will be described: steroidogenesis, LH receptors, and changes in levels of inhibin/activin and IGF/IGFBP.

Extensive reviews covering these compounds have been published recently for sheep and cattle (9, 76, 136). The main steps of follicular maturation are outlined on Figure 3.

The maturational step associated with follicle recruitment is the appearance of aromatase activity within the granulosa layer. Size at recruitment and size at which aromatase first becomes detectable closely coincides in sheep (2 mm) and in cattle (3 to 4 mm). This is a key maturational step because it enables the follicle to produce estradiol from androgen precursors produced by theca cells. Most follicles in the cohort also have the ability to produce activin and inhibin, since the mRNAs coding for the alpha and both beta subunits are already present at this stage of development (73). The steep increase in FSH secretion which follows cauterization of all follicles over 5 mm in diameter (57) strongly suggests that they are inhibin dominated. Follistatin is produced by cohort follicles (118) in amounts which increase with increasing follicle size. As a consequence, more and more activin is bound to follistatin as the follicle enlarges and follicles become more and more heavily inhibin dominated.

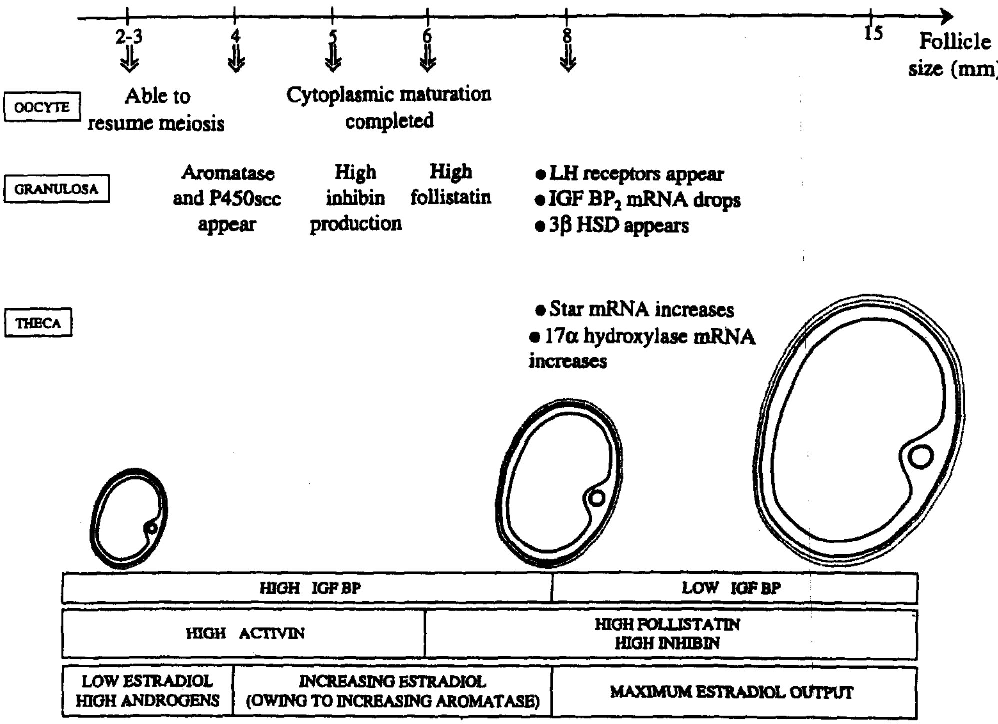

Figure 3. Summary of the main steps of oocyte, granulosa and theca cell differentiation occurring in cattle as the follicles develop through gonadotropin-dependent folliculogenesis. Consequences for follicular maturation are presented on the bottom part of the figure.

In cattle, there are three key maturational steps associated with selection and appearance of a dominant follicle. First, the appearance of LH receptors on granulosa cells (75) is a pre-requisite for establishment of follicular dominance (see below) and ovulation following an LH surge. Second, the reduction in the amounts of IGF binding proteins such as IGFBP2 and IGFBP4 (35, 86, 87) appears to be mediated by reduced production (IGFBP2) or increased proteolysis (IGFBP4). Both occur when the follicle is around 8 mm in diameter. Third, there is a selective decrease in the amounts of the 34kD inhibin dimer present in follicular fluid, while no major change is detectable amongst the other inhibin forms (86).

There are two key steps in oocyte maturation: nuclear maturation when the oocyte resumes meiosis to reach the Metaphase 2 stage and cytoplasmic maturation where the oocyte gains the ability to be fertilized and develop to the blastocyst stage thereafter. The average follicle sizes at which the oocyte acquires the ability to undergo nuclear maturation (2 to 3 mm) and at which cytoplasmic maturation is completed (over 6 mm) have been established in cattle (53, 80). Once again, there is wide variation providing evidence that the relationship between follicle size and oocyte function are complex. Beneficial effects on cytoplasmic maturation of two groups of follicular compounds (EGF/TGF alpha and inhibin/activin) have been identified (48, 129). In contrast, analysis for individual follicles, of the links between follicular maturation and oocyte quality has failed to identify many correlations between markers of follicular function and fate of the oocyte such as fertilization, development to the blastocyst stage (48).

ENDOCRINE AND LOCAL CONTROL OF TURNOVER OF GONADOTROPIN-DEPENDENT FOLLICLES

These mechanisms are well documented in cattle, therefore this section will specifically deal with this species. Reference to other species will only be used when data in cattle are not available. This section will attempt to answer the following 4 questions: 1) What are the mechanisms involved in the control of recruitment? 2) What are the mechanisms involved in selection of the ovulatory follicle? 3) Why does the dominant follicle grow while other follicles regress? and 4) What causes the dominant follicle to turn-over during successive anovulatory waves? Potential endocrine regulators include FSH and LH while the discussion of the role of local regulators will be limited to compounds whose concentrations change throughout follicular maturation (see above).

What are the Mechanisms Controlling Recruitment?

Three types of evidence have conclusively demonstrated that FSH is the key hormone inducing recruitment (Figure 4). An association between a FSH surge and recruitment has been demonstrated in cattle (65) and sheep (64) at several physiological stages. This temporal relationship has been strengthened by other observations in the following two types of experimental paradigms: In the first, models were developed in which endogenous FSH secretion is blocked and exogenous FSH is injected (33, 136). While the presence of decreased concentrations of FSH resulted in the disappearance of follicles larger than 4 mm in diameter, injection of pure FSH reinitiated recruitment. In the second, the timing of the FSH surge or its magnitude/duration were manipulated in cyclic females by treatment with steroid-free follicular fluid resulting in a postponement of the recruitment of the new follicular wave (132).

While there is general agreement that FSH is the key regulator of recruitment in most species in sheep (98) and swine (45), the action of FSH on the ovary appears to be complex. There is a minimal threshold of FSH concentration below which recruitment cannot proceed. This minimum threshold appears to be variable between animals (97) but stable throughout time for a specific human subject (27). In addition, large variations in FSH thresholds between follicles in a specific ewe have also been demonstrated (55). Figure 4 illustrates how these two types of thresholds interact to generate an ovulatory follicle from the cohort of developing follicles. The main effects of FSH are to induce aromatase activity within granulosa cells (111), explaining why follicles gain the ability to produce estradiol, stimulate the production of inhibin and follistatin (76,118) and initiate the repression of the gene coding for IGFBP2 (5).

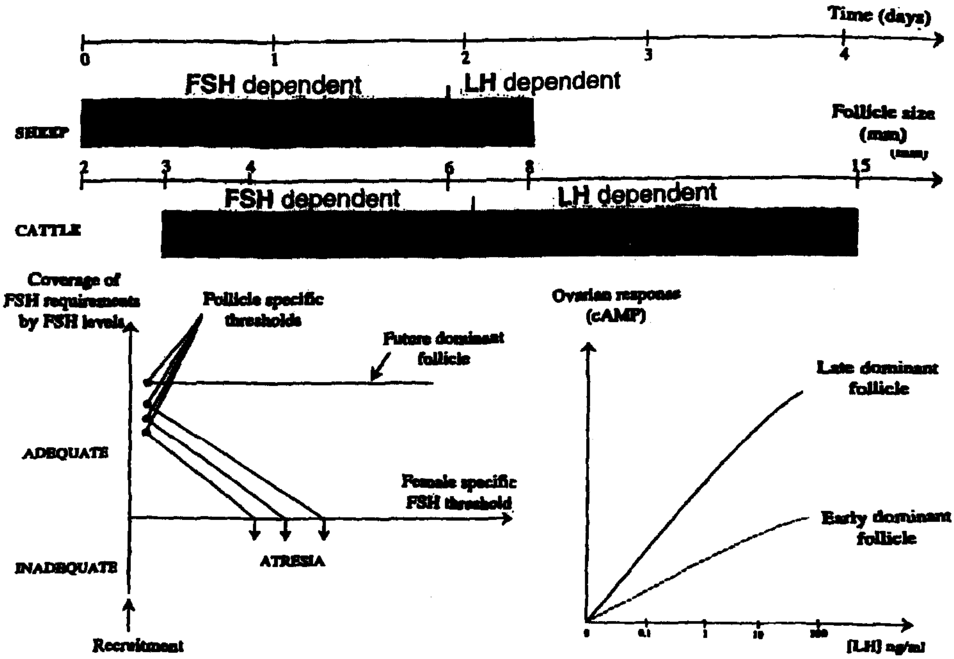

Figure 4. Gonadotropin requirements of sheep and cattle follicles according to their size (top panel). On the bottom panel, the mechanisms modulating follicle response to FSH (left graph) or LH (right graph) are presented. Ovarian response to FSH appears to operate through thresholds which are specific for each female and follicle. Ovarian response to LH is primarily modulated by the stage of follicular dominance and associated changes in receptors for LH

LH appears to be minimally involved in the control of recruitment as evidenced by the observation that recruitment proceeds similarly in a number of endocrine environments in which LH pulses frequency are reduced (early-luteal phase), low (mid-luteal phase) or very low (prepubertal stage, postpartum anestrus) (Figure 5)(51,89,114,120). Two locally produced compounds may also modulate recruitment: IGF's and follistatin. The effect of the former has been demonstrated following administration of exogenous growth hormone which resulted in increased circulating IGF1 concentrations, and increased numbers of recruited follicle per wave (66). Active immunization against follistatin also resulted in an increased number of follicles in the cohort (119). Hence, follistatin may limit and IGF1 may increase the size of the cohort of recruited follicles.

What are the Mechanisms Involved in the Selection of the Ovulatory Follicle?

The selection process has been extensively studied and two theories are generally offered to explain the mechanisms involved. In the first, selection is controlled by endocrine mechanisms only (reduction in FSH support to follicles) and in the second, there is production of compounds by the largest follicle which directly inhibit development of the other follicles of the cohort. Since both theories are not mutually exclusive, it is likely that in species with a loose control of ovulation rate (sheep), only the FSH dependent mechanism operates, while in species with a strict control of ovulation rate (cattle, primates) both mechanisms may be operating.

The idea that the drop in FSH concentrations, which occur 2 to 3 days after recruitment, is a key mechanism in the selection process, is supported by several lines of evidence. First, in most species, there is a close temporal relationship between selection and the time when FSH levels reach their nadir (Figure 5). This drop in FSH support is caused by the combined action of inhibin and estradiol which are produced by follicles > 5 mm in diameter and act by negative feed back on the pituitary (57, 86). Second, an injection of exogenous FSH to delay the drop in FSH results in a delay in the selection process in cattle (1, 86) and sheep (49). Finally, in GnRH antagonist-treated ewes, a reduction in FSH results in atresia of all cohort follicles (29), conclusively demonstrating that a deprivation of FSH can trigger atresia. Follicles with the highest FSH requirement will be hit first, ensuring the fine tuning of the size of the cohort to the normal number of ovulations for that species (Figure 4).

There are, however, data which suggest that the FSH-operated mechanism is not the only one involved in selection. First, in experiments using GnRH immunized cows (33), it was difficult to obtain single ovulations by only manipulating FSH and LH concentrations. This may indicate that an additional regulatory step (independent of gonadotropin concentrations) is a key regulator of the selection process. Second, data from superovulation trials show that ovarian responses to gonadotropins injected when a dominant follicle is present are depressed (1,72,74,138) indicating that large follicles can reduce sensitivity to gonadotrophins of smaller ones. Direct inhibitory effects of follicular fluid on follicular growth and maturation have been confirmed by in vivo and in vitro approaches. The inhibitory effects observed after injection of inhibin-free follicular fluid (78) or after injection of follicular fluid in inhibin immunized cows (139) can be taken as evidence confirming the presence of bio-active compounds in follicular fluid. In vitro approaches have been used to identify aromatase as a target for the effects of follicular fluid (30, 44, 109). Candidate compounds mediating these effects could be TGF alpha, inhibin, superoxyde dismutase or heat shock protein 90 since all four have the ability to reduce basal or FSH-induced aromatase activity. Whether these compounds are preferentially produced by dominant follicles, and whether some or all are actively secreted in ovarian venous serum (to reach the contralateral ovary) has not yet been established.

What are the Mechanisms Involved in the Final Growth of the Dominant Follicle?

There is general agreement that LH is the key hormone involved in the final growth of the dominant follicle while other follicles in the cohort complete atresia (Figure 4). The approaches used to support this claim are numerous. They include the pioneering studies demonstrating that regression of the corpus luteum in cows treated with CIDR-B devices was associated with an increased pulsatility of LH and the extension of the dominance phase (121, 128). In addition, exogenous injection of LH pulses to post-partum beef cows, in which a limited pulsatile pattern of LH secretion has been demonstrated, results in an increased maximum diameter of the dominant follicles and increased duration of dominance (50). In underfed beef cows, a low frequency of LH pulses was associated with a reduction in the size of the dominant follicle and a reduced interwave interval resulting in a preponderance of 3-wave cycles (71). Furthermore, treatment of cows with a GnRH agonist, which acutely suppresses LH pulses, blocks follicular growth at the size at which follicles become dominant i.e., 8 to 9 mm (136). Finally, in GnRH-immunized cows supplemented with a regime of exogenous gonadotropins, administration of LH is required once follicles have reached 8 mm in diameter to grow to an ovulatory size (33). This last observation agrees with data demonstrating the presence of the message coding for the LH receptor (141) or the LH receptor itself (75) in granulosa cells as soon as the follicles reach 8mm in diameter. Hence, in cattle, the development of LH receptors on granulosa cells closely coincides with onset of follicular dominance. The hormonal balance required for a successful transition to a LH-dependent stage has been recently characterized in sheep (29), and may also apply for cattle. It involves a progressive decrease in FSH concentrations combined with high pulsatile secretion of LH (29).

Two local factors may be re-enforcing follicular dominance, namely IGF1 and VEGF. This is supported by the observation that in GH-deficient cattle, which display a limited IGF1 production, follicular dominance does not occur and follicles are arrested at 8 mm in diameter (31). This indicates that LH receptor induction might be, at least partially, mediated by IGF1. In addition, LH has been shown to stimulate the production of vascular endothelium growth factor (VEGF), a potent stimulator of angiogenesis (56) and IGF1 may re-enforce LH action on angiogenesis. Overall, an increase in blood supply to the dominant follicle will maximize gonadotrophin delivery. Indeed, evidence for selective delivery of gonadotrophins to the dominant follicle is available in primates (142).

In contrast, estradiol production does not appear to be a pre-requisite for follicular dominance. This is supported by at least three lines of evidence. First, treatment of ewes (18, 94) or primates (143) with aromatase inhibitors, which markedly reduce estradiol output, during the mid-follicular phase does not block the development of dominant follicles. Second, the administration of epostane, a steroid synthesis inhibitor, successfully decreases steroid output by large follicles, but does not prevent the development of dominant follicles (135); the number of such follicles is even increased compared to the control group. Third, dominant follicles produce variable amounts of estradiol according to the physiological status of the female. This ranges from low in follicles developing during the post-partum period, to limited during the first wave of the luteal phase, and high for those developing during the follicular phase. Despite these differences, the magnitude of dominance, measured by the size gap between the dominant and largest subordinate follicle, is similar.

What causes Dominant Follicle Turnover?

Dominant follicles are exquisitely sensitive to LH, thus it is no surprise that changes in the pattern of pulsatile LH secretion may alter their fate, hence inducing wave turnover. Solid evidence supporting this claim has been presented, using the persistent follicle model (128). Demise of the dominant follicle could be induced by a doubling of circulating progesterone concentrations, which resulted in a 50% reduction in the frequency of LH pulses. This claim is in good agreement with the observed time of turnover of the first wave DF (D7 to 10) which coincides with peak progesterone concentrations and a minimal frequency of LH pulses (Figure 5). The physiological consequences of LH starvation have not been fully clarified. Some studies (7) report that androgen output collapses before a decrease in aromatase can be observed. Others (52) indicate that aromatase is the first steroidogenic enzyme to vanish when dominant follicles turn-over.

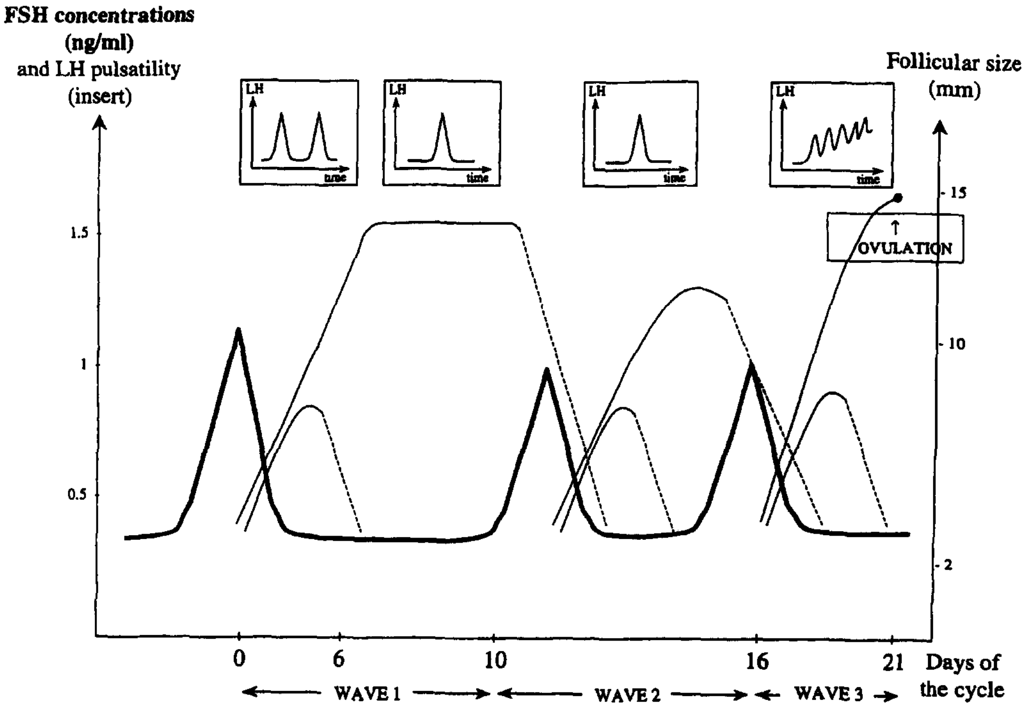

Figure 5. Changes in FSH concentrations and in the frequency of LH pulses seen as waves develop during estrous cycle of cattle. Note that recruitment is associated with high FSH concentrations, that selection occurs when FSH concentrations reach their nadir and that wave turn-over occurs when LH pulse frequency becomes minimal. Maximal diameter of the dominant follicle of Wave 2 is also reduced due to the low LH pulse frequency at this stage.

ENDOCRINE AND LOCAL CONTROL OF MATURATION OF GONADOTROPIN-DEPENDENT FOLLICLES

This section will be restricted to the control of steroidogenesis, as most information is available in this area. Other reviews have covered the control mechanisms of the other steps of follicular maturation, such as differentiation of LH receptors on granulosa cells (136), or cytoplasmic maturation of the oocyte (48). In Figure 6 are summarized the key steps in the regulation of follicular steroidogenesis. In follicles 3 to 8 mm in diameter, FSH is the primary stimulating gonadotrophin; after binding to its receptors on granulosa cells, FSH causes the induction of aromatase activity (111). Granulosa cells then become capable of producing estradiol when provided with an androgen precursor. FSH also stimulates follistatin output (118) which binds follicular activin contributing to the change in the inhibin/activin balance towards inhibin. Finally, FSH reduces the expression of the message coding for IGFBP2 (4,5). This results in increased amounts of free IGF1 which has the ability to synergize with FSH to further increase aromatase activity.

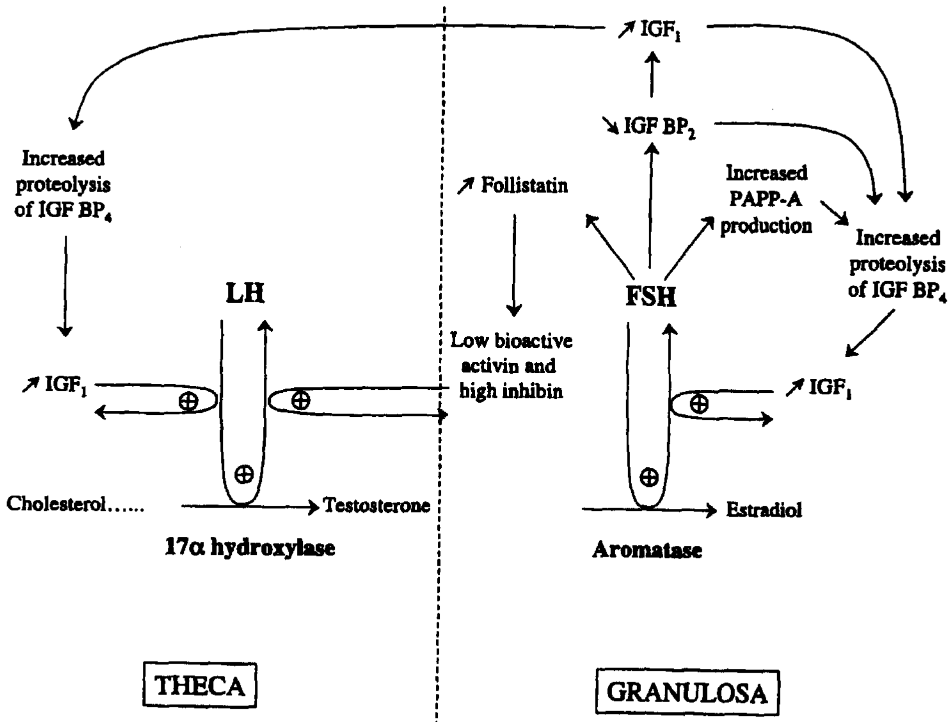

Figure 6. Mechanisms involved in the development of an estrogen-active dominant follicle. Before selection, FSH reduces amounts of IGFBP2, and stimulates production of follistatin and a protease (PAPP-A) degrading IGFBP4. The net results of FSH action are increased inhibin and bio-available IGF1 within the follicle. IGF1 synergizes with FSH to induce aromatase. During the dominance phase, the main gonadotropin becomes LH whose main action is to stimulate androgen output by theca cells. The increased amounts of IGF1 generated by increased proteolysis of IGFBP4 potentiates the action of LH. A paracrine effect of inhibin produced within the granulosa cell layer also potentiates the action of LH.

In follicles more than 8 mm in diameter, the key gonadotrophin that allows development to continue is LH. Its main effect is to stimulate androgen output by theca cells. Inhibin, produced in large amounts by granulosa cells may also exert a stimulatory paracrine action on androgen production (140). In addition, the low amounts of activin, together with the large amounts of free IGF1 may result in increased proteolysis of IGFBP4 (83). This will result in increased bio-availabilities of IGF1 and IGF2 at the level of granulosa and theca cells, respectively. Within the theca layer, the IGF's synergize with LH to increase androgen production further. Within the granulosa layer, the IGF's will increase cell sensitivity to LH further. As a consequence, expression of P450 SCC will be upregulated and 3 BHSD will appear in granulosa cells (9). Overall, the induction of aromatase in 3 to 8 mm follicles followed by the increasing provision of progestagen/androgen substrate when follicles reach sizes larger than 8 mm results in follicles with a fully functional steroidogenetic mechanism.

TURNOVER OF GONADOTROPIN-DEPENDENT FOLLICLES: PRACTICAL CONSEQUENCES FOR ESTRUS CONTROL TECHNIQUES

The pre-requisites for high fertility are a short interval between AI and ovulation and optimal conditions for the embryo to be fertilized, develop, and prevent luteolysis. Estrus control techniques should ensure synchronous ovulation (enabling the use of fixed-time AI), by manipulating follicular wave patterns prior to the removal of the suppressive effects of progesterone/progestagens on LH release. This will be briefly described in the following section. The possible impact of the manipulation of luteal phase follicular waves on fertility will then be discussed.

Estrus Control Treatments to Optimize Fertility of the Ovulatory Follicle

The target for a successful estrus synchronization treatment is precise control of onset of estrus (which will allow fixed-time AI without estrus detection) together with high fertility at the synchronized estrus (at least 55 % of females pregnant). The target animals are beef cattle which are generally non-cyclic at the time of treatment (at least under French conditions) and dairy cattle which are generally cyclic around 50 days post partum. A recent report indicates, however, that high milk production is associated with a reduced proportion of normally cycling animals at the beginning of the breeding period (92). To date, estrus control strategies are mainly based on controlling the life span of the corpus luteum with prostaglandins, or induction of ovulation with GnRH or its agonists, or prevention of estrus using progestagen treatments. Prostaglandins alone do not provide acceptable synchrony, because the time to ovulation depends on the stage of development of the dominant follicle at the time of prostaglandin-induced luteolysis. Because of this, a treatment combining prostaglandins and GnRH (GPG) has been developed which provides a greater degree of synchrony in estrus and ovulation. The initial GnRH injection causes ovulation or luteinisation of the large follicle present on the ovary and synchronizes the recruitment of a new follicular wave. At 7 days following GnRH injection, an injection of prostaglandin induces regression of the corpus luteum and allows for final maturation of the synchronized dominant follicle. A second GnRH injection, given 48h after prostaglandin injection, synchronizes the time of ovulation of the dominant follicle. Progestagen implants, progesterone-impregnated vaginal devices or oral progestagens are flexible tools, to treat populations of mixed (cyclic and non cyclic) females, a situation commonly found in farm practice. A small amount of PMSG (generally 400 to 500 IU) is generally injected at the end of treatment to induce ovulation in non-cyclic females and induce a tighter synchrony in cyclic females. Optimization of the use of progesterone or progestagens (Norgestomet, MAP, MGA) must ensure that the treatment period is shorter than the duration of a luteal phase of the cycle and that an additional luteolytic/antiluteotrophic treatment is administered. Estradiol benzoate or valerate is normally administered at the time of progestagen treatment to initiate uterine-induced luteolysis. When injected early after ovulation, estradiol appears to generally be antiluteotrophic (79). When injected in the presence of an active corpus luteum, estradiol is generally luteolytic, although its ability to induce corpus luteum regression is more limited early in its life phase (Days 3-5 post ovulation) than later on (101). Failure of estradiol-induced CL regression will result in lack of synchrony. An alternative is to inject prostaglandin at the end of progestagen treatment as the CL present at this time should be sensitive to prostaglandins. As with prostaglandin treatment, the stage of development of the dominant follicle at progestagen removal is closely related to the time interval to estrus and ovulation. If the treatment does not synchronize follicular wave development, it is likely that a wide range of follicular sizes (and stages of dominant follicle development) will exist at end of the progestagen treatment resulting in poor synchronization of ovulation. This suggests that treatments should be refined to improve control of the time of wave emergence, in order to provide for a suitable stage of dominant follicle development at the time progestagen is removed.

Efficient estrus synchronization treatment protocols need to induce atresia of all large follicles present on the ovaries whatever their developmental stage, resulting in recruitment of a new follicular wave, synchronous development of a new dominant follicle in all females, and ovulation at a predictable time. Recently acquired knowledge on the specific gonadotropin requirements of follicles at specific stages of development (see above) has helped to understand the actions of the following compounds used for estrus synchronisation.

Progesterone/progestagens, when given alone, suppress LH in a dose-dependent fashion. Low doses of progesterone reduce maximal size of the dominant follicle by about 3 mm (19). High doses of progesterone alone can induce dominant follicle turnover, a finding clearly demonstrated in the persistent dominant follicle model (121). The time to follicle turnover appears somewhat variable, as it depends on the stage of the dominant follicle and on the total amount of progesterone (endogenous + exogenous) circulating. Administration of progestagens in the presence of a functional CL, before selection of a dominant follicle, will allow selection to occur, but DF will be smaller and regress earlier. On the other hand, if low doses of progestagens are administered, as is normally done in estrus synchronization programs and the CL regresses during treatment, the DF present at that time will persist and continue to grow.

GnRH induces LH release and ovulation of LH-dependent follicles (i.e., those more than 8 mm in diameter) which will result in the emergence of a new follicular wave. However, GnRH/LH will not induce ovulation of smaller FSH-dependent follicles and as a result, follicular wave emergence will not be synchronized by a single GnRH injection in all animals.

Estradiol has a range of effects depending on the dose and the compound used (estradiol-17B, estradiol valerate or benzoate)(23, 24, 25). Overall, when injected in the presence of a FSH dependent follicle, it induces its regression by reducing FSH concentrations through negative feed-back. New wave emergence will occur when estradiol levels return to baseline (at a time which varies with the ester used) and a new FSH rise occurs. When injected in the presence of a LH-dependent follicle, estradiol does not seem to induce follicle turn-over but prevents wave emergence until it has once again reached baseline. Estradiol can be most successfully used in combination with progestagens. The rationale for this association is as follows. Estradiol causes turn-over of FSH-dependent follicles, while progestagens will induce turn-over of LH-dependent follicles. Regardless of the stage of the follicular wave at the time of initiation of treatment, combined treatment with estradiol-17B and progestagen successfully induced new wave emergence approximately 4 days later (22). This is the basis for most of the commercial synchronization treatments presently on the market.

Reduced fertility following AI at synchronized estrus could be the consequence of either an inappropriate time interval between AI and ovulation (see previous section) or of ovulation of an aged oocyte into a suboptimal oviductal environment (21,104). The latter has been shown to occur when progestagen treatments were initiated late in the estrous cycle, when the CL had regressed spontaneously during treatment. The LH pulse frequency observed after luteal regression and under progestagen treatment is then intermediate between that of luteal and follicular phases resulting in the development of a persistent dominant follicle (121, 128), which will contain an aged oocyte (which may have resumed meiosis) (104). In addition, the persistent dominant follicle will produce compounds (estradiol?) which may alter the pattern of expression of oviductal proteins (21). Successful synchronization treatment protocols involving the use of progesterone/progestagens will only be successful if they prevent the development of a persistent dominant follicle.

Management of the Luteal Follicular Waves to Raise Fertility

Since progesterone prepares the uterine environment to permit embryo development, elevated progesterone concentrations may improve embryo survival. This could be achieved by induction of ovulation (via hCG or GnRH) of the dominant follicle that develops after AI, which will result in the formation of accessory corpora lutea, or by providing exogenous proegesterone (CIDR insertion) following ovulation. In one study involving cows with limited fertility, hCG administration between Days 5 and 7 following estrus did improve fertility by 15% (117). In addition, treatment of repeat breeder cows with hCG at Day 5.5 after ovulation increased conception rate (134). However, in another study, the administration of hCG had no effect on animals of high and low (summer heat stress) fertility (115). In contrast to the rather positive effects of hCG administration, there is no evidence that CIDR insertion post ovulation has any beneficial effects on fertility (133).

Treatment with a GnRH agonist between Days 11 and 12 following insemination has been shown to increase pregnancy rates (81,82,105). The underlying mechanism is presumed to be that luteinization or ovulation of the dominant follicle present at that time, would decrease estradiol concentrations and postpone luteolysis. Efficacy of this type of treatment may be related to the stage of dominant follicle development at the time of treatment since no effect can be expected if no dominant follicle is present and to the side of dominant follicle development relative to the side of the pregnant uterine horn since little effect can be expected if the dominant follicle developing during the late luteal phase is contralateral to the uterine horn carrying the young embryo.

MOBILISATION OF GONADOTROPIN-SENSITIVE FOLLICLES FOR SUPEROVULATION : POSSIBILITIES AND LIMITS

Although alternative techniques exist, such as ovum pick-up, combined with in vitro maturation, fertilization and development, superovulation remains an efficient and inexpensive procedure to maximize the number of young born from females of high genetic merit. For superovulation, pure FSH (sheep), pituitary extracts containing FSH and also LH (cattle) or PMSG/eCG (all species) with or without the use of antisera to PMSG are generally used (37). Owing to its long half-life, PMSG is simple to administer (as opposed to the twice-daily injection required when using FSH) generating less stress on the animals and being less time-consuming. Because the ovarian response to both gonadotropins is dose dependent and highly variable, it is difficult to demonstrate whether one type of treatment is superior to another. In the following section, insights on the features of follicular growth and maturation following treatment with FSH or PMSG will be provided. Techniques which may be useful to maximize ovulation rate, decrease its variability or predict the ovarian response, will then be explained.

Follicular Growth and Maturation in Gonadotropin Treated Females

Ovarian stimulation with PMSG or FSH results in increased numbers of large growing follicles, of which 70% generally ovulate. Ovulation is spread over 24 to 48 h. Increased ovulation rate is obtained through a combination of 5 mechanisms. A decrease in the size at which follicles are recruited (to 0.8 mm and 1.3 mm in sheep and cattle), and a decrease in the size of the preovulatory follicles by about 3 mm are generally observed in superovulated cattle. In addition, superovulation doubles follicular growth rates which reach 3 mm/day (47). Finally, the prevention of atresia and extension of recruitment, even in the presence of already large follicles allow more follicles to be included in the ovulatory cohort. When exogenous gonadotropins are injected, recruitment can be increased or extended while selection is blocked, events which are in agreement with our knowledge of the mechanisms controlling these events in cyclic animals (see above). The mean number of growing follicles generally increases with the amount of gonadotropin injected until a maximum is reached, which is closely related to the number of follicles present in the ovaries of the treated females (88). The numbers of such gonadotropin-responsive follicles varies 20 fold between females. In any case, owing to the steady accumulation of follicles which are all included in the pool of ovulatory follicles, there is a huge heterogeneity in the morphological features of the ovulatory follicles in superstimulated animals. Follicle size and/or the number of granulosa cells per follicle within the pool of superstimulated follicles in a specific animal may vary 3- to 4-fold (42).

It may, however, not be realistic to try to reach each female's maximum response because massive ovarian responses are often associated with a number of problems. First, it should be remembered that the final goal of superovulation is to obtain good quality embryos. Large ovarian responses often result in reduced embryo quality. Second, massive ovarian responses may be associated with a reduction in the proportion of follicles that ovulate (112). This is probably the consequence of a reduction in follicle responsiveness to LH (112) and/or of the production of the gonadotropin surge inhibiting factor (Gnsif) by the extra large cohort of growing follicles (54).

Follicular maturation is also strongly altered when superovulation is induced. The extensive studies of Price's group in cattle (122,123,124) have characterized the alterations in differentiation of granulosa/thecal cells induced by PMSG or FSH. Follicles whose growth is induced by PMSG contain reduced amounts of FSH and LH receptors. In contrast, the amounts of mRNA coding for Star, 17-alpha hydroxylase and aromatase are higher when PMSG rather than FSH is used. The up-regulation of the first of these enzymes is likely to be responsible for the increased progesterone output of PMSG stimulated follicles (42) since amounts of P450scc and 3 beta HSD were not altered by treatment (123, 124). Whether these high progesterone concentrations have detrimental effects on oocyte quality remains to be established. A large functional heterogeneity among all follicles in a superovulated cohort is also obvious. The possible links between a specific steroid/peptide environment and oocyte quality in a given follicle are worth further investigations (using a combination of OPU and FIV/MIV/DIV).

Variability in Gonadotropin Induced Ovulation Rate

Two aspects of variability are relevant to the embryo transfer industry, namely variability between females and within each female. While the first can be detected as soon as superovulation is attempted, the second requires successive attempts at superovulation in the same female, an elite cow in most cases. Our understanding of follicular dynamics has helped us to clarify the mechanisms involved in each case.

Between females. It was demonstrated more than 15 years ago, (88) that there was a tight relationship between follicular populations within a specific animal and its superovulatory response to gonadotropin treatments. This observation underlines the interest in techniques whereby follicular populations can be estimated prior to treatment. A direct estimation of the numbers of follicles in the emerging cohort can be obtained in cattle by ultrasonography (108), although it is difficult to accurately count follicles smaller than 3mm in diameter. In any case, good and poor responders can be identified, starting at Day 3 following injection of PMSG in cattle and at estrus in sheep and goats (47). In contrast, because of the smaller size of the follicles mobilized by exogenous gonadotropin treatments in sheep/goats, the ultrasonic approach prior to treatment is unlikely to be feasible or accurate. A possible alternative in predicting superovulatory responses in sheep could be the measurement of inhibin A concentrations (68). In all cases, the maximum ovarian response will only be achieved when superovulation is attempted under optimal conditions (i.e., in the absence of a dominant follicle; see below).

Within females. When a specific female is repeatedly superovulated, the objective is to obtain a constant response to repeated attempts, as close as possible as the ovarian maximum for that female. Such an objective implies that this female is always treated under optimum conditions in the absence of a dominant follicle. There are numerous lines of evidence demonstrating that growing dominant follicles have detrimental effects on the mobilization of smaller follicles stimulated to grow by the injection of exogenous gonadotropins (1, 28, 72, 74, 138). The larger the dominant follicle at the time of treatment; the lower is the superstimulation obtained (100). In addition, the presence of a growing dominant follicle may interfere with the ability of gonadotropin stimulated follicles to ovulate (72, 128). Treatment in the absence of a growing dominant follicle is difficult to perform because of the unpredictability of follicular wave patterns (2 or 3 waves) and the necessity to constantly monitor prospective donors with ultrasonography. Novel approaches (cautery of the large follicle, steroid administration) have been tested in cattle and proved to be similarly efficient (10). Another option is to treat very early in the cycle, at the time of recruitment of Wave 1 (2). In sheep and goats, control of the wave pattern by GnRH agonists/antagonists has produced promising results to overcome this problem (32), although the negative effects of the presence of a dominant follicle appear more limited than in cattle (49).

When females are repeatedly treated with nonhomologous gonadotropins, antibodies may be generated. While there is little evidence supporting this in cattle, in goats, a fraction of the females repeatedly treated with non-homologous gonadotropins developed antibodies which reduced or prevented successful superovulation at later treatments (103). Recent studies (110) have demonstrated a major genetic component for the immune response against exogenous gonadotropins. While this finding does not resolve the problem, it is helpful, since in animals likely to quickly develop antibodies OPU/FIV may be used instead of superovulation.

Maximizing Ovulation Rate

Two strategies available to embryo transfer practitioners are to increase the population of gonadotropin-responsive follicles, or to increase the duration of gonadotropin stimulation in order to mobilize more small antral follicles. However, it should be remembered that these approaches will not control variability and are even likely to increase it.

Some treatments that have been shown to have additive effects with exogenous gonadotropins include immunization against follistatin (119), steroids (96), follicular fluid (91) and treatment with growth hormone (67). Since the benefits of these extra treatments are generally not obvious because the total number of embryos, or the numbers of transferable embryos are not regularly increased, they have not gained widespread use. One possible bonus for the use of such treatments is that they may allow the use of reduced amounts of exogenous hormones to obtain a specific level of ovarian response.

Treatments which extend the duration of ovarian stimulation have recently been proposed (11 38, 93). They rely on a GnRH agonist/antagonist block of endogenous gonadotropins release prior to ovulation and provision of an exogenously-induced LH surge to allow an extension of the duration of the follicular phase. This extra time is believed to allow more follicles to be recruited and more time for all follicles in the cohort to grow and mature prior to the ovulatory release of LH. It is also believed not to be detrimental to larger follicles that are already present. However, this idea has recently been challenged, at least for sheep (93). An added bonus is that the time of ovulation is precisely known, which should facilitate synchrony between AI and ovulation (11). Although promising results have been reported (38), it remains to be shown that clear beneficial effects on the number of transferable embryos can be obtained.

CONCLUSION

Our understanding of follicular growth and maturation has considerably improved in the past ten years. Although the wave pattern of follicular growth is now well understood, it is still not clear why the ovary generates waves to optimize its function. In induced ovulators such as the camel, living in remote environments, a wave-like pattern without spontaneous ovulation would maximize the chance of becoming pregnant as follicles are available for ovulation at almost any time. However, the context is vastly different in farm animals. A working hypothesis could be that follicular waves, at least during the luteal phase, are a pre-requisite for luteolysis to occur. Estradiol is involved in the luteolytic cascade in ruminants (16) and the wave mechanism could generate the amounts of estradiol required for luteolysis. The wave mechanism would then be a way to prepare a new follicle for ovulation following luteolysis. Indirect evidence supporting this hypothesis is that luteolysis operates independently of follicular estrogens in swine (16) and no luteal waves of follicular growth are detectable in this species.

The development of new treatment protocols, possibly derived from the physiological signals described in this review, would certainly improve the efficacy of procedures for estrus control and help to optimize superovulatory reponse to gonadotropin treatments. The control of ovarian variability and the analysis of its underlying mechanisms is the major challenge for scientists during the next few years.

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